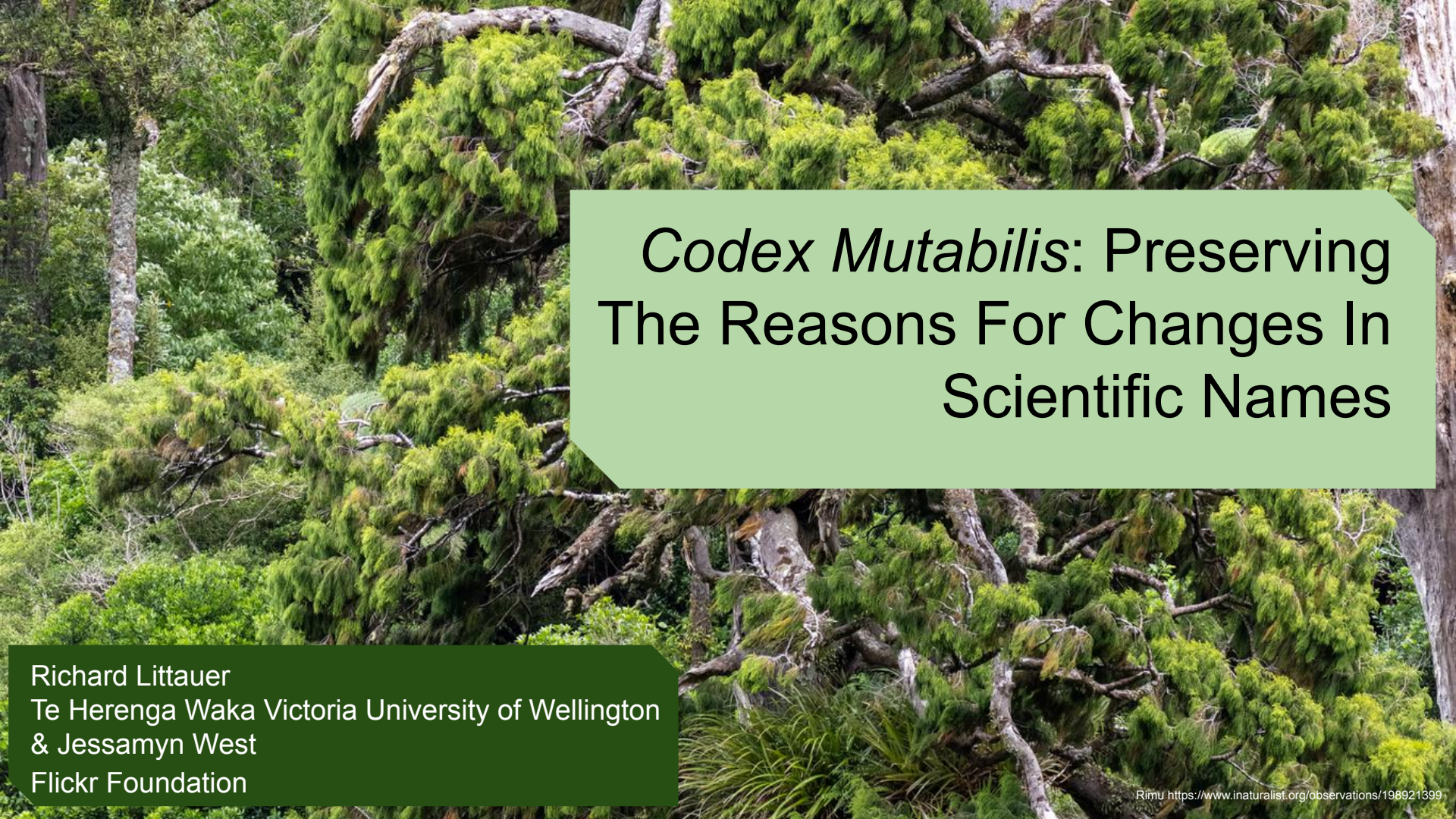
# *Codex Mutabilis*: Preserving The Reasons For Changes In Scientific Names

Richard Littauer
Te Herenga Waka Victoria University of Wellington
& Jessamyn West
Flickr Foundation



Rimu https://www.inaturalist.org/observations/198921399

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# INTERNATIONAL CODE OF ZOOLOGICAL NOMENCLATURE

**Fourth Edition**

[Incorporating **A.** Declaration 45, Addition of Recommendations to Article 73 and of the term "specimen, preserved" to the Glossary, **B.** Declaration 44, amendments of Article 74.7.3, with effect from 31 December 1999, and **C.** the Amendment on e-publication, amendments to Articles 8, 9, 10, 21 and 78, with effect from 1 January 2012]

***adopted by the***
***International Union of Biological Sciences***

The provisions of this Code supersede those of the previous editions with effect from 1 January 2000




ISBN 0 85301 006 4

The author of this Code is the International Commission on Zoological Nomenclature

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# INTERNATIONAL COD...

**Fourth Edition**

[Incorporating **A.** Declaration 45, Addition of ... 44, amendments of Article 74.7.3, with effect... 21 and 78, with effect from 1 January 2012]

***adopted by the International Union of Biological Sciences***

The provisions of this Code supersede those ...

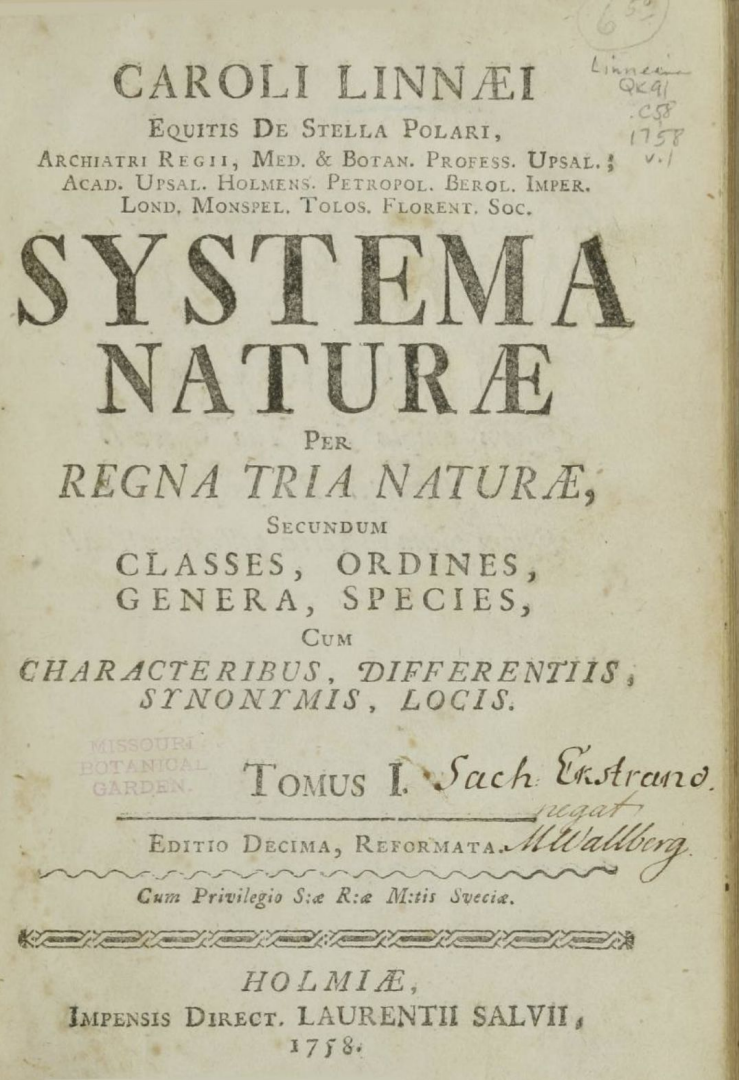
ICZN

ISBN 0 85301 006 4

The author of this Code is the International Commission on Zoological Nomenclature

## "The Code":

- French and English
- 90 Articles
- Glossary
- Examples (which don't count)



CAROLI LINNÆI

EQUITIS DE STELLA POLARI,

ARCHIATRI REGII, MED. & BOTAN. PROFESS. UPSAL.;
ACAD. UPSAL. HOLMENS. PETROPOL. BEROL. IMPER.
LOND. MONSPEL. TOLOS. FLORENT. SOC.

# SYSTEMA NATURÆ

PER

*REGNA TRIA NATURÆ,*

SECUNDUM

CLASSES, ORDINES,
GENERA, SPECIES,

CUM

*CHARACTERIBUS, DIFFERENTIIS,*
*SYNONYMIS, LOCIS.*

TOMUS I.

EDITIO DECIMA, REFORMATA.

*Cum Privilegio S:æ R:æ M:tis Sveciæ.*

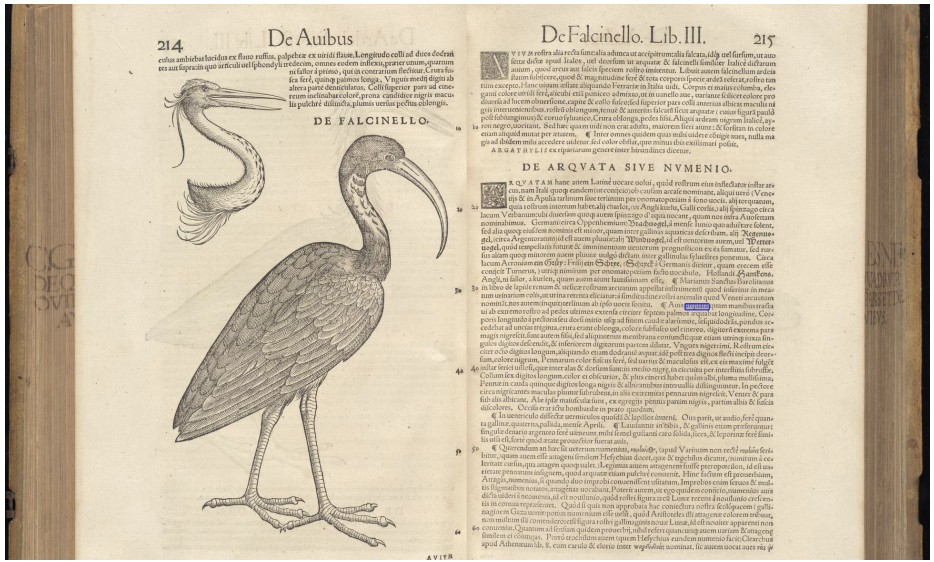
HOLMIÆ,

IMPENSIS DIRECT. LAURENTII SALVII,

1758.

# Article 3: The beginning

- Starts with Linnaeus 1758, on Jan. 1.
- But the Code doesn't specify which calendar.



214 De Auibus

DE FALCINELLO.

De Falcinello. Lib. III. 215

DE ARQVATA SIVE NVMENIO.

Linnaeus https://www.biodiversitylibrary.org/page/727037#page/3/mode/1up
Gessner: https://archive.org/details/gri_33125010867691/page/214/mode/2up?q=Arquata

# Article 11.2 Latin alphabet

*Haematopus unicolor*

And not:




© Sebastian Doak https://www.inaturalist.org/photos/107051293

# Article 11.3 Derivation

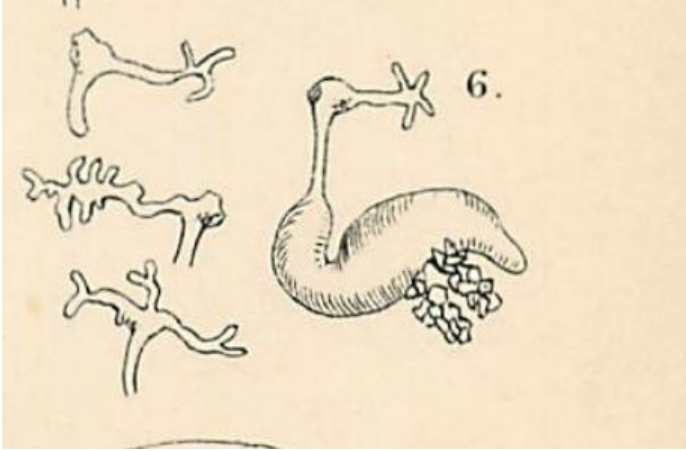
Doesn't have to be Latin.

Cf. *Manu antiquus* Marples, 1946



© BJ Marples for the Royal Society of New Zealand - https://paperspast.natlib.govt.nz/periodicals/TPRSNZ1946-76.2.9.8

# Article 11.9 Species names




Can be:

- an adjective in the nominative singular
- a noun in the nominative singular
- a noun in the genitive case, or
- an adjective in the genitive case representing related species

Bassett-Smith, P.W. (1896). Notes on the parasitic Copepoda of fish obtained at Plymouth, with descriptions of new species. Annals and Magazine of Natural History (6)18:8-16, pls. 3-6. (vii-1896)
https://www.marinespecies.org/aphia.php?p=sourcedetails&id=71842

# Article 23 Principle of priority

First means first.



Gould, public domain

- †***Heteralocha acutirostris*** (Gould) **Huia**

  *Neomorpha acutirostris* Gould, 1837: *Synop. Birds Australia 1*: pl. 11 & text – North Island.

  *Neomorpha crassirostris* Gould, 1837: *Synop. Birds Australia 1*: pl. 11 & text – North Island.

  *Neomorpha Gouldii* G.R. Gray, 1841: *List Gen. Birds* (2nd edition): 15. Unnecessary *nomen novum* for *Neomorpha acutirostris* Gould, 1837.

  *Heteralocha Gouldi* (G.R. Gray); Cabanis 1851, *Mus. Heineanum 1*: 218. Unjustified emendation.

  *Heteralocha gouldi* (G.R. Gray); G.R. Gray 1862, *Ibis 4*: 217. Unjustified emendation.

  *Heteralocha acutirostris* (Gould); Buller 1872 (Apr.), *History of the Birds of N.Z.*, 1st edition (part 1): 63.

  *Heteralocha auctirostris* (Gould); Enys 1876, *Trans. Proc. N.Z. Inst. 8*: 204. Misspelling.

## LES MOHOUAS (2).

Ils tiennent des picucules, des grimpereaux et des gobe-mouches. Leur bec est gros, pointu, courbé, à peine infléchi à sa pointe, qui est légèrement arrondie, et pourvue d'une très petite dentelure. Les narines sont ouvertes et médianes; la mandibule supérieure est carénée, et l'inférieure est rectiligne. La commissure du bec est garnie de soies roides et longues. La langue est ciliée. Les ailes s'allongent jusqu'au milieu de la queue. Celle-ci a douze pennes fortes, étagées, et usées à leur sommet. Les tarses sont très forts.

La seule espèce de ce groupe a été découverte sur les bords de la baie Tasman, où les naturels lui donnent le nom de *mohoua-houa* : c'est le *grimpereau hétéroclite* (3) à dos olivâtre, à ailes brunâtres frangées de jaunâtre, et tête, poitrine et ventre jaunes. Son gésier ne renfermoit que des baies.



(1) *Sclerurus*, ibid.

(2) *Mohoua*, Less.

(3) *Certhia heteroclites*

fig. 1, p. 223

Lesson 1837 (?): Oeuvres completes de Buffon, 9 p. 139

# Article 24 First Reviser

- *Pachyptila belcheri* vs. *Pseudoprion turtur solanderi* Mathews, 1912 – Mathews
- *Larus dominicanus antipodum* vs. *antipodus* Bruch, 1853
- *Branta* vs. *Brandae* Scopoli, 1769
- *Mohoua, Mohouas, Mohoua-houa* Lesson, 183x?
- *Rhopalum succineicollarum, succineicolle, suecineicolle,* or *succinei collearum* [and *succineicollare*] Tsuneki, 1952

# Article 26. Assumption of Greek or Latin



Badly transliterated:

- Athene, Gerygone, Eudynamys, Aegotheles, Gymnorhina

Latin or Greek?

- Lagopus, Lago+pus?, λαγοπους?

© Christoph Moning https://www.inaturalist.org/observations/113797579

# Article 30.1 Gender of genus-group names

If Latin, Latin.

If Greek, Greek.

If Greek but looks like Latin, like Latin.

If disclaimed, not Greek or Latin.

If “common or variable”, masculine.

If ends in -*ops*, masculine.



# Article 30.2 Gender of genus-group names

If in “modern European language”,

the gender of that word.

Else, as noted by author.

Or by combination.

Or masculine;

unless ends in -*a* (feminine);

-*um*, -*on*, or -*u* (neuter).



# Article 31.2 Agreement

Species must agree with genus.

31.2.1: But not compound names.

31.2.2: Unclear N or Adj:

- Did the author note it?
- Is usage decisive?
- Can it be regarded as either?



# Some unclear names from NZ I've found

*Megalapteryx didinus, Euryapteryx curtus, Euryapteryx curtus curtus, Dendrocygna, Cygnus atratus, Cygnus sumnerensis chathamicus, Branta, Numida, Callipepla, Callipepla californica, Synoicus, Ptilinopus, Eudynamys, Aegotheles, Hirundapus, Hirundapus caudacutus, Hirundapus caudacutus caudacutus, Apus, Apus pacificus, Apus pacificus pacificus, Zapornia pusilla, Tribonyx hodgenorum, Haematopus, Pluvialis dominicus, Numenius hudsonicus, Arenaria, Calidris, Calidris acuminata, Calidris ferruginea, Calidris alpina, Calidris melanotos, Calidris pusilla, Phalaropus, Phalaropus fulicarius, Phalaropus lobatus, Anous albivittus, Gygis alba, Larus dominicanus dominicanus antipodum, Onychoprion fuscatus serratus, Onychoprion anaethetus melanopterus, Chlidonias leucopterus, Chlidonias hybridus, Chlidonias hybridus javanicus, Aptenodytes patagonicus, Pygoscelis papua taeniata, Diomedea, Diomedea antipodensis, Diomedea antipodensis antipodensis, Thalassarche, Thalassarche impavida, Hydrobates leucorhous, Thalassoica, Daption capense, Ardenna pacifica chlororhyncha, Pelecanoides georgicus, Pelecanoides georgicus georgicus, Morus, Sula leucogaster plotus, Bubulcus ibis coromandus, Ardea intermedia plumifera, Botaurus poiciloptilus, Aquila, Tyto alba delicatula, Athene, Eurystomus orientalis pacificus, Todiramphus, Todiramphus sanctus, Cacatua galerita, Platycercus eximius, Cyanoramphus, Cyanoramphus novaezelandiae chathamensis, Anthornis, Anthornis melanura obscura, Anthochaera carunculata, Gerygone, Gerygone albofrontata, Mohoua, Artamus cyanopterus, Gymnorhina, Petroica, Petroica macrocephala, Petroica macrocephala macrocephala, Poodytes punctatus stewartianus, Passer domesticus, Acanthis, Acanthis flammea, Anas, Oreortyx, Coturnix coromandelica, Lophura nycthemera, Lagopus, Lagopus lagopus scotica, Pluvialis apricaria, Passer montanus, Alectoris rufa*

# The Code has some issues

Here's a cute parrot again.

- ***Strigops habroptila*** G.R. Gray

  *Strigops habroptilus* G.R. Gray, 1845: *Gen. Bird.*
  *Strigopsis habroptilus* (G.R. Gray); Bonaparte 18
  *Strigops greyii* G.R. Gray, 1862: *Ibis 4*: 230 – So
  *Stringops* (*Strigops*) *habroptilus* (G.R. Gray); Po
  *Stringops habroptilus* G.R. Gray; Reichenow 188
  *Stringops Greyi* G.R. Gray; Reichenow 1881, *Jo*
  *Strigops habroptilus habroptilus* G.R. Gray; Mat
  *Strigops habroptilus innominatus* Mathews & Ire
  *Strigops habroptilus parsonsi* Mathews & Iredal
  *Strigops habroptila* G.R. Gray; Dickinson 2003,



# The Code has some issues

- No storage of decisions
- Papers often behind paywalls
- The Code rarely updates
- Some changes are too small

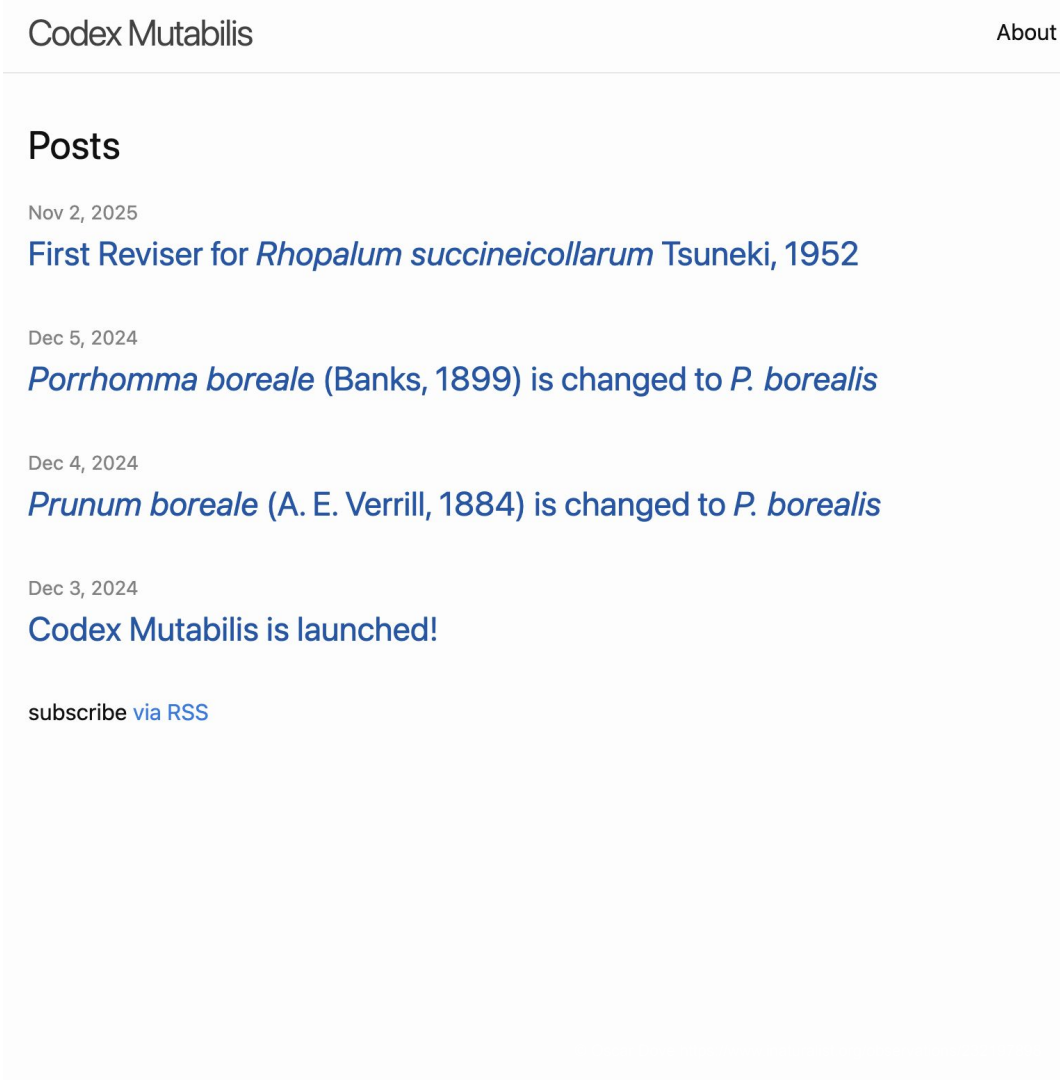
This causes:

- Instability for names
- Duplicated work
- Confusion



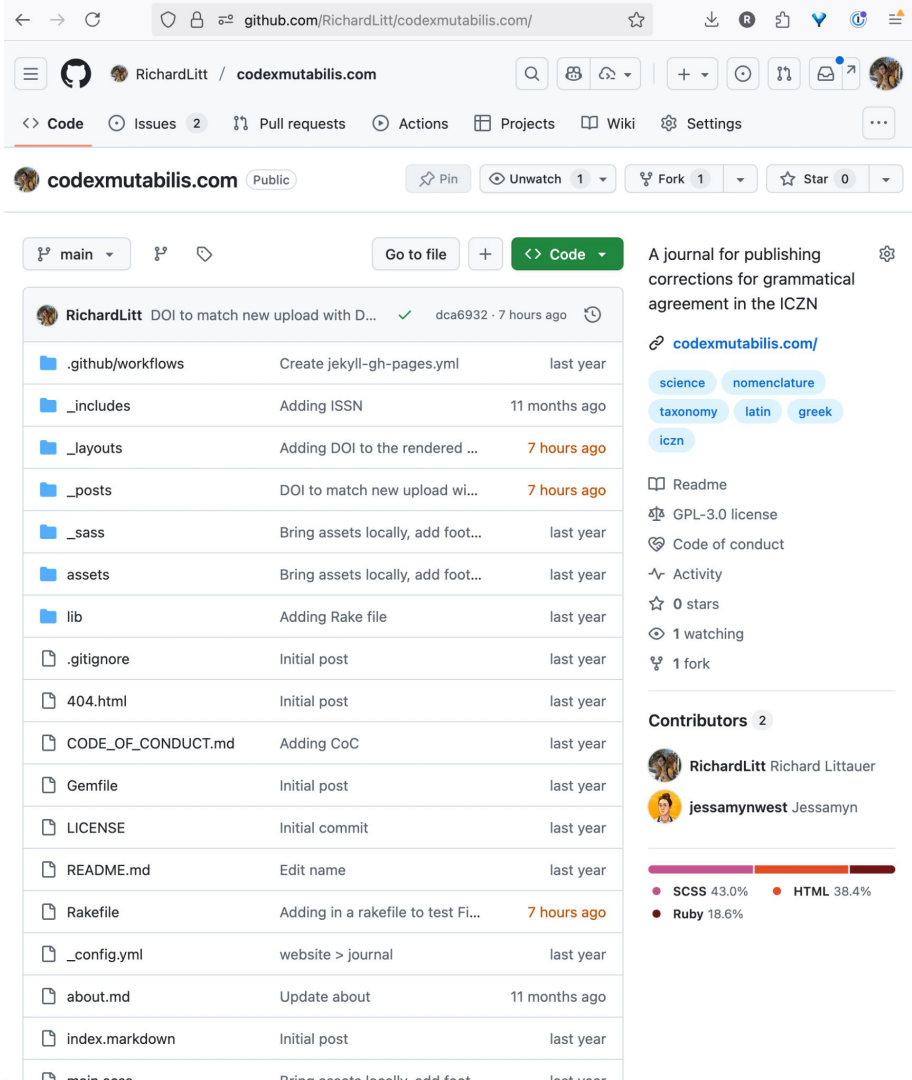
# Codex Mutabilis

- Satisfies the ICZN rules around publishing online
  - Permanent works (PDFs)
  - Storage (Internet Archive)
  - An ISSN
- Provides a place to make:
  - Mandatory changes
  - First revisions
  - Synonymies



# Codex Mutabilis

- A GitHub Jekyll website
- Publishes in Markdown
- Uses GitHub pull-requests for review
- Backed up on the Internet Archive
- Low maintenance costs



# Each article has:

- A list of relevant Articles in the Code
- References
- Etymologies
- Editorial or peer review
- A machine-readable interface
- Version control
- Persistent identifiers (DOI and Git commits)
- Registry in Zoobank
- Persistent storage of the data.



## First Reviser for *Rhopalum succineicollarum* Tsuneki, 1952

Nov 2, 2025 · Richard Littauer · https://doi.org/10.5281/zenodo.17504393

*Rhopalum succineicollarum* Tsuneki, 1952 was made available according to the International Code of Zoological Nomenclature (ICZN 1999; henceforth "the Code") in a publication that included the spellings *succineicollarum* (pg. 115, 116, 120), *succineicolle* or *suecineicolle* (pg. 116), and *succinei collearum* [sic] (pg. 117). A synonymy and citation record is provided in Leclercq (2010). Tsuneki consistently used *succineicollare* in subsequent publications, but this form is not a valid mandatory emendation of *collarum*, *collearum*, or *colle* according to the rules of Latin grammar. Tsuneki cannot be considered his own First Reviser according to Article 24.2.4 of the Code.

Haneda (2008) may have served as a First Reviser, as according to Leclercq (2010) the form *succineicollarum* was used in this work. However, this work is not accessible online, and I could not verify whether Haneda (2008) cited all forms of the original name in Tsuneki (1952). Hashimoto and Nakanishi (1999) also use *succineicollarum*, but do not act as First Revisers.

Acting as First Reviser, I select *Rhopalum succineicollarum* Tsuneki, 1952 as the available name. *Rhopalum succineicolle*, *Rhopalum suecineicolle*, *Rhopalum succineicollearum*, and *Rhopalum succineicollare* are all incorrect spellings and therefore unavailable according to Art. 24.2.3. If they are considered subsequent spellings, both Haneda (2008) and Hashimoto and Nakanishi (1999) preclude the possibility of *Rhopalum succineicollare* being a correct original spelling according to Art. 33.3.1. If they are to be considered incorrect original spellings, they have no separate availability and cannot enter into homonymy or be used as a substitute name according to Art. 32.4.

If Haneda (2008) proves to have been a First Reviser, this revision is to be considered nullified according to Art. 24.2.5 and an update will be published accordingly upon notice.

### References

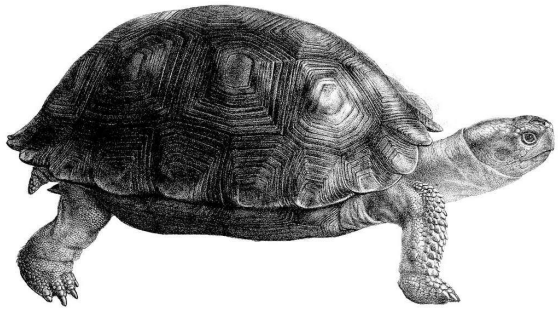

- Haneda, Y. (2008). Taxonomic guide to the Japanese Aculeate Hymenoptera. 8. Subfamily Crabroninae (1). Tsunekibachi 13, p. 1-17.
- Hashimoto, Y. and Nakanishi, A. (1999). Additions to the list of Tsuneki's type-specimens of Hymenoptera in the Museum of Nature and Human Activites, Hyogo. Nature and Human Activities 4, p. 55-58. Available from: https://www.hitohaku.jp/publication/r-bulletin/Nature%20and%20Human%20Activities%20No.04%201999%20055-058%20optimized.pdf
- ICZN (1999). International Code of Zoological Nomenclature. 4th Edition. International Trust for Zoological Nomenclature, London, 306 pp. Available from: iczn.org.
- Leclercq, J. (2010). Hyménoptères Crabroniens d'Asie du genre Rhopalum Stephens 1829 (Hymenoptera: Crabronidae: Crabroninae). Entomologie faunistique, 63(1), p. 3-21. Available from: https://popups.uliege.be/2030-6318/index.php?id=1507&file=1&pid=1487.
- Tsuneki, K. (1952). The genus Rhopalum Kirby (1829) of Japan, Korea, Saghalien and the Kuriles, with a suggested reclassification of the subgenus and descriptions of four new species (Hym. Sphecidae, Crabroninae). Journal of the Faculty of Science, Hokkaido University, series VI, Zoology 11(1), p. 110-125. Available from: https://eprints.lib.hokudai.ac.jp/dspace/bitstream/2115/27118/1/11(1)_P110-125.pdf


### Registration

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# Why?

- The Code is too slow.
- Don’t just store decisions! Store the rationale behind them.
- Don’t lock things in PDFs.
- Experiments can show the limits of a system, thereby improving it.



Proceedings of the Zoological Society of London - Testudo Elephantopus. Public Domain.

# Needs for Codex Mutabilis



# Questions?

Thank you.

Reach me at http://richard.social, or

Codexmutabilis.com 👇



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